$$\int u \cdot v'\, dx = u \cdot v - \int u' \cdot v\, dx$$

$$\int u dv = uv - \int v du$$

1) $$\int \left(x^2+x\right)e^{-x}\, dx = \left\{\begin{matrix} u=\left(x^2+x\right) \\ v'=e^{-x} \\ v=-e^{-x} \end{matrix}\right\} = -\left(x^2+x\right)e^{-x} + \int \left(2x+1\right)\cdot e^{-x}\, dx =$$

$$= \left\{\begin{matrix} u=\left(2x+1\right) \\ v'=e^{-x} \\ v=-e^{-x} \end{matrix}\right\} = -\left(x^2+x\right)e^{-x} - \left(2x+1\right)e^{-x} + 2\int e^{-x}\, dx =$$

$$= -\left(x^2+x\right)e^{-x} - \left(2x+1\right)e^{-x} - 2e^{-x} + C = -e^{-x}\left(x^2+x+2x+1+2\right) + C =$$

$$= -e^{-x}\left(x^2+3x+3\right) + C$$

2) $$\int \left(7-2x\right)\cdot \sin x\, dx = \left\{\begin{matrix} u=\left(7-2x\right) \\ v'=sinx \\ v=-cosx \end{matrix}\right\} = -\left(7-2x\right)\cdot cosx - \int \left(-2\right)\cdot\left(-cosx\right) dx =$$

$$= \left(2x-7\right)\cdot cosx - \int 2cosx\, dx = \left(2x-7\right)\cdot cosx - 2sinx + C$$

3) $$I = \int e^x \cdot \sin x\, dx = \left\{\begin{matrix} u=sinx \\ v'=e^x \\ v=e^x \end{matrix}\right\} = sinx\cdot e^x - \int cosx\cdot e^x\, dx = \left\{\begin{matrix} u=\cos x \\ v'=e^x \\ v=e^x \end{matrix}\right\} =$$

$$= sinx\cdot e^x - \left(cosx\cdot e^x + \int sinx\, e^x\, dx\right) = sinx\cdot e^x - cosx\cdot e^x - \int sinx\, e^x\, dx$$

$$I = sinx\cdot e^x - cosx\cdot e^x - I$$

$$2I = sinx\cdot e^x - cosx\cdot e^x + C$$

$$I = \frac{1}{2}e^x\left(sinx - cosx\right) + C$$

Другая версия

$$I = \int e^x \cdot \sin x\, dx = \left\{\begin{matrix} u=e^x \\ v'=sinx \\ v=-cosx \end{matrix}\right\} = -cosx\, e^x + \int e^x\cdot cosx\, dx = \left\{\begin{matrix} u=e^x \\ v'=cosx \\ v=sinx \end{matrix}\right\} =$$

$$= -cosx\, e^x + e^x\, sinx - \int e^x\cdot sinx\, dx$$

$$I = -cosx\, e^x + e^x\, sinx - I$$

$$2I = e^x\, sinx - cosx\, e^x$$

$$I = \frac{1}{2}e^x\left(sinx - cosx\right) + C$$

3) $$\int_{\frac{\pi}{2}}^{3\pi} 4x\cdot cos(2x)\, dx = \left\{\begin{matrix} u=4x \\ v'=cos\left(2x\right) \\ v=0.5\, sin\left(2x\right) \end{matrix}\right\} = \left(4x\cdot 0.5\, sin\left(2x\right)\right)\Big|_{\frac{\pi}{2}}^{3\pi} - \int_{\frac{\pi}{2}}^{3\pi} 4\cdot 0.5\, sin\left(2x\right) dx =$$

$$= 2x\, sin\left(2x\right)\Big|_{\frac{\pi}{2}}^{3\pi} - \int_{\frac{\pi}{2}}^{3\pi} 2\, sin\left(2x\right) dx = \left(2x\, sin\left(2x\right) + \frac{2cos\left(2x\right)}{2}\right)\Big|_{\frac{\pi}{2}}^{3\pi} =$$

$$= \left(2x\, sin\left(2x\right) + cos\left(2x\right)\right)\Big|_{\frac{\pi}{2}}^{3\pi} = \left(2\cdot 3\pi\, sin\left(2\cdot 3\pi\right) + cos\left(2\cdot 3\pi\right)\right) - \left(2\cdot\frac{\pi}{2}\, sin\left(2\cdot\frac{\pi}{2}\right) + cos\left(2\cdot\frac{\pi}{2}\right)\right) =$$

$$= \left(0+1\right) - \left(0-1\right) = 1-\left(-1\right) = 2$$

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